A common number regularization for Brownian clock calculus: full predictable limits and correction duality

Project Stardust
A formalization companion

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Abstract

Let W be a two-sided Brownian scenery independent of a Brownian clock B. For a real Schwartz function g, spatially smooth versions of $g(a + W_{|B_t|}) - g(a)$ have a genuine scenery-valued reflected Itô formula. We prove that one fixed injective smoothing of the clock Gaussian number operator gives a strong limit of the entire predictable integrand in the original time–clock–scenery L^2 space. The full curvature and local-time correction converges in the associated common dual norm. The limits are independent of the fixed nonnegative compact C^3 mass-one profile, and hold as its positive width tends continuously to zero. The decisive bound is proportional to $t^{-1/2}$ and is integrable at the origin. Fixed clock sectors converge in ordinary joint L^2, while independent rectangular source/clock truncations have uniformly small tails in the dual norm. Every statement retains the original Brownian-clock filtration and all scenery variables. No unsmoothed correction limit or intrinsic observed-filtration IBM Itô formula is asserted. Classical Gaussian spectral and Hilbert-valued stochastic mechanisms are credited; research priority for this particular assembly remains unresolved.

1 The original spaces and the result

Fix independent standard Brownian sources W and B, with W two-sided and $B_0 = 0$. Thus

$$\mathbb{E}_W(W_r - W_s)^2 = |r - s|, \qquad \mathbb{E}_B B_t^2 = t.$$

The reflected-clock observation is $Z_t = W_{|B_t|}$. Work with real spaces

$$E = L^2(\Omega_W), \qquad \mathcal{H} = L^2(\Omega_B; E) \simeq L^2(\Omega_B \times \Omega_W).$$

All these are the original sources, with their full Gaussian sigma algebras. Fubini identifications are isometries and do not replace the joint variables by variables with matching laws. A scenery-valued predictable process means a strongly measurable E-valued process predictable for the Brownian clock filtration. Its space at horizon $T \geq 0$ is the closed Hilbert space

$$\mathcal{D}_T = L^2_{\mathrm{pred}}((0, T] \times \Omega_B; E).$$

The original Hilbert-valued Itô integral is the isometry

$$\mathsf{J}_T : \mathcal{D}_T \longrightarrow \mathcal{H}, \qquad \mathsf{J}_T u = \int_0^T u_t \, dB_t, \qquad \|\mathsf{J}_T u\|_{\mathcal{H}} = \|u\|_{\mathcal{D}_T}. \tag{1}$$

The scenery is the value-space parameter; it is not averaged out. Equivalently one may reveal the whole independent scenery at time zero and integrate in B. This convention is different from predictability for the smaller observed past $\sigma(Z_s : s \le t)$.

Let P_k denote the orthogonal projection onto the kth clock Wiener chaos, tensored with the identity on E. Put

$$\mathsf{S} = P_0 + N_B^{-1}, \qquad N_B^{-1} f = \sum_{k\ge 1} \frac{1}{k} P_k f. \tag{2}$$

The inverse here is zero on the clock vacuum. The added P_0 is essential. In particular S is an injective self-adjoint contraction on $\mathcal{H}$, retaining all scenery chaoses.

Fix $g \in \mathcal{S}(\mathbb{R};\mathbb{R})$ and $a \in \mathbb{R}$. A profile is a fixed function $\rho \in C_c^3(\mathbb{R})$ such that $\rho \ge 0$ and $\int \rho = 1$. Evenness is not required. For $\varepsilon > 0$ set

$$\begin{aligned} \rho_\varepsilon(x) &= \varepsilon^{-1}\rho(x/\varepsilon), \\ S_\varepsilon(r) &= \int_{\mathbb{R}} \rho_\varepsilon(r-y) W_y \, dy \quad \text{in } E, \qquad (3) \\ h_\varepsilon(r) &= g(a + S_\varepsilon(r)), \\ F_{\varepsilon,t} &= h_\varepsilon(|B_t|) - h_\varepsilon(0), \qquad F_{0,t} = g(a+Z_t) - g(a). \qquad (4) \end{aligned}$$

The integral in (3) is a Bochner convolution. The same S_ε is used at both endpoint arguments. The random $h_\varepsilon(0)$ generally differs from $g(a)$.

The raw integrand and raw correction are

$$U_{\varepsilon,t} = \operatorname{sgn}(B_t) h_\varepsilon'(|B_t|), \tag{5}$$

$$C_{\varepsilon,T} = L_T^0(B) h_\varepsilon'(0) + \frac{1}{2}\int_0^T h_\varepsilon''(|B_t|)\, dt. \tag{6}$$

Local time has the Tanaka convention $|B_T| = \int_0^T \operatorname{sgn}(B_t)\, dB_t + L_T^0(B)$. For each positive width these are actual elements of $\mathcal{D}_T$ and $\mathcal{H}$, respectively. Define the bounded predictable operator

$$\mathsf{A}_T u = \sum_{k\ge 0} \frac{1}{k+1} \Pi_k u, \tag{7}$$

where Π_k is clock-chaos projection on the genuine time sections. Its construction, predictability and integral intertwining are proved below.

Theorem 1.1 (Full common regularized limit)**.** *For every g, a, T as above there are $u_{0,T} \in \mathcal{D}_T$ and $c_{0,T} \in \mathcal{H}$, independent of the fixed admissible profile, such that as $\varepsilon \downarrow 0$ through all positive real widths,*

$$\mathsf{A}_T U_\varepsilon \longrightarrow u_{0,T} \qquad \text{in } \mathcal{D}_T, \tag{8}$$

$$\mathsf{S} C_{\varepsilon,T} \longrightarrow c_{0,T} \qquad \text{in } \mathcal{H}, \tag{9}$$

$$\mathsf{S} F_{0,T} = \mathsf{J}_T u_{0,T} + c_{0,T}. \tag{10}$$

For $t > 0$, with $v_t = 1_{(0,t]}/\sqrt{t}$ in the clock Cameron–Martin space, define $R_v = \langle v, D_B N_B^{-1} \cdot \rangle$. The time sections of the limit are

$$u_{0,T}(t) = t^{-1/2} R_{v_t} F_{0,t} \quad \text{for almost every } 0 < t \le T. \tag{11}$$

For $L = \sup|g'|$ and $m_1 = \mathbb{E}|G|$ for a standard real Gaussian,

$$\|u_{0,T}(t)\|_{\mathcal{H}}^2 \le L^2 m_1 t^{-1/2}, \qquad \|u_{0,T}\|_{\mathcal{D}_T}^2 \le 2L^2 m_1 \sqrt{T}. \tag{12}$$

The same section bound holds for $\mathsf{A}_T U_\varepsilon$, uniformly in the width and the admissible profile.

For $T = 0$ all spaces, integrals and endpoint differences in this assertion are zero. No statement about a common exceptional sample set for all terminal times is needed: this is a theorem at each fixed horizon.

Verified scope. The frozen source is commit `bab688bb6140cd6316c80680095a914ea0d346db`. The accompanying theorem map records exact declaration headers, source hashes and permanent line links. The principal endpoint, predictable limit and dual-closure modules have been reviewed with their proofs. The existing full-build and complete transitive axiom-audit receipts are checked by hash; this editorial paper changes no Lean source and does not claim a new library build. Gaussian chaos completeness, the constructed original Brownian Itô isometry, and its established Tanaka/local-time formula are prerequisites. We reconstruct their application and the convergence proof, rather than relabeling those prerequisites as new theorems.

2 Spatial smoothing without a hidden profile constant

The map $r \mapsto W_r$ is continuous into E, with exact increment norm $\|W_r - W_s\|_E = \sqrt{|r-s|}$. Hence the convolution in (3) exists on a compact interval for every r. Changing variables gives the useful identity

$$S_\varepsilon(r) = \int_{\mathbb{R}} \rho(z) W_{r-\varepsilon z}\, dz.$$

This change of variables also fixes the sign convention for nonsymmetric profiles.

Lemma 2.1 (Uniform increment and approximation bounds)**.** *Suppose* $\operatorname{supp}\rho \subset [-R, R]$, $R \geq 0$. *Then*

$$\|S_\varepsilon(r) - S_\varepsilon(s)\|_E \leq \sqrt{|r-s|}, \tag{13}$$

$$\|S_\varepsilon(r) - W_r\|_E^2 \leq \varepsilon R, \tag{14}$$

$$\|h_\varepsilon(r) - h_\varepsilon(s)\|_E \leq L\sqrt{|r-s|}. \tag{15}$$

In particular $h_\varepsilon(r) \to g(a + W_r)$ *in* E *at every fixed* r.

Proof. Minkowski's integral inequality, positivity of ρ and its unit mass give

$$\Big\| \int \rho(z)(W_{r-\varepsilon z} - W_{s-\varepsilon z})\, dz \Big\|_E \leq \int \rho(z)\sqrt{|r-s|}\, dz = \sqrt{|r-s|}.$$

For the approximation subtract $W_r \int \rho(z)\, dz$ inside the Bochner integral and use $\sqrt{\varepsilon|z|} \leq \sqrt{\varepsilon R}$. The ordinary mean value theorem gives $|g(a+x) - g(a+y)| \leq L|x-y|$. Apply it to the genuine jointly Gaussian scenery variables and take their L^2 norm. This proves (15) and the last assertion. □

Neither derivative norms of ρ_ε nor a bound on R enters (13) or (15). Radius R occurs only in the approximation estimate. The proof uses a fixed admissible profile; it does not assert convergence for arbitrary width-dependent profiles with unbounded supports or unproved approximation properties.

Lemma 2.2 (Endpoint convergence and energy)**.** *For every fixed* $t \geq 0$, $F_{\varepsilon,t} \to F_{0,t}$ *in* $\mathcal{H}$. *For all positive widths,*

$$\|F_{\varepsilon,t}\|_{\mathcal{H}}^2 \leq L^2 m_1 \sqrt{t}, \qquad \|F_{0,t}\|_{\mathcal{H}}^2 \leq L^2 m_1 \sqrt{t}. \tag{16}$$

Proof. For every deterministic r, Lemma 2.1 gives $h_\varepsilon(r) - h_\varepsilon(0) \to g(a + W_r) - g(a)$ in E. Both differences have E norm at most $2 \sup |g|$. Integrating their squared difference against the law of $|B_t|$ proves joint L^2 convergence by dominated convergence. Alternatively the uniform scenery error gives a quantitative endpoint bound by two applications of the Lipschitz inequality. For the energy, first hold the clock fixed and use (15) with $s = 0$. Its squared norm is at most $L^2|B_t|$. Since $\mathbb{E}|B_t| = m_1\sqrt{t}$, Fubini proves the first bound. Take the norm limit for the second. At $t = 0$ the endpoint differences are identically zero. □

Source correspondences:
`IBMContinuousScenery.sourceScenery_increment_sq`,
`IBMDilatedMollifier.scenery_error`,
`IBMCompactKernelIncrements.spatial_increment`,
`IBMCompactClockGraph.endpoint_uniform_energy`, and
`IBMDilatedEndpointLimit.endpoint_continuum_tendsto`.

3 The genuine finite-width reflected formula

Finite-width regularity is used before taking any singular limit. Differentiation of the compact convolution in E gives S'_ε and S''_ε. Gaussian moment estimates and bounded derivatives of g justify the E-valued chain rules

$$h'_\varepsilon(r) = g'(a + S_\varepsilon(r))S'_\varepsilon(r), \tag{17}$$
$$h''_\varepsilon(r) = g''(a + S_\varepsilon(r))(S'_\varepsilon(r))^2 + g'(a + S_\varepsilon(r))S''_\varepsilon(r). \tag{18}$$

Here the products are actual scenery variables. They are not defined by assigning Gaussian coefficients after the fact. Indeed the spatial derivative of smoothed Brownian scenery has variance $\|\rho_\varepsilon\|_2^2$, and the second derivative has variance $\|\rho'_\varepsilon\|_2^2$. Thus globally in r,

$$\|h'_\varepsilon(r)\|_E \le L\|\rho_\varepsilon\|_2,$$
$$\|h''_\varepsilon(r)\|_E \le 2 \sup |g''| \, \|\rho_\varepsilon\|_2^2 + L\|\rho'_\varepsilon\|_2.$$

The factor 2 bounds the square root of the Gaussian fourth moment factor 3. Compact C^3 regularity ensures the required continuity of the second derivative. To justify the chain rule directly, use Gaussian L^p bounds for the convolution difference quotients for each finite p, Taylor's formula with the bounded derivatives of g, and Hölder's inequality for the remainder products. Their limits in L^2 give (17)–(18).

Proposition 3.1 (Original finite-width identity)**.** *For every $\varepsilon > 0$, $U_\varepsilon \in \mathcal{D}_T$, $C_{\varepsilon,T} \in \mathcal{H}$, and*

$$F_{\varepsilon,T} = \mathsf{J}_T U_\varepsilon + C_{\varepsilon,T}. \tag{19}$$

Proof. Continuity of the E-valued slope and Brownian predictability show that (5) has a predictable version. The preceding global slope bound makes its time L^2 norm finite. The curvature term in (6) is a pathwise Bochner time integral of a continuous E-valued function and has deterministically bounded E norm for this fixed width. The boundary term belongs to $\mathcal{H}$ because $h'_\varepsilon(0) \in E$ and $L^0_T(B) \in L^2(\Omega_B)$.

For any continuous real linear functional ℓ on E, apply the scalar C^2 Itô formula to $\ell(h_\varepsilon(|B_t|))$. Tanaka gives $d|B_t| = \operatorname{sgn}(B_t)dB_t + dL^0_t$; its quadratic variation is dt, since the zero set has Lebesgue measure zero almost surely. Local-time mass is supported on $B_t = 0$, so the supported Stieltjes

integral is $\ell(h'_\varepsilon(0))L^0_T$. The remaining finite-variation term is exactly half the curvature integral. All scalarizations commute with the original Hilbert-valued Itô and Bochner integrals. This proves (19) after every ℓ. Choose a countable separating family in the separable Hilbert space E and intersect its full-probability sets to recover the E-valued identity. It is an equality in $\mathcal{H}$. □

In particular (6) constructs the correction before (19) is used. The expression is not merely a name for an unspecified residual. Individual derivative bounds here grow as the width shrinks, so they do not prove convergence of either raw term.
Source correspondences:
`IBMCompactKernelSchwartz.spatial_hasDerivAt`,
`IBMCompactKernelSchwartz.slope_hasDerivAt`, and the actual integrand, whole correction and finite identity in
`IBMCompactKernelReflectedIto`. The constructed Tanaka and supported-local-time theorem is the prerequisite also used in the sticky-clock and one-time-law companions [3].

4 Clock number smoothing and predictable degree shift

We give the operator mechanisms explicitly. Orthogonality yields $f = \sum_{k\geq 0} P_k f$ strongly in $\mathcal{H}$. With

$$w_0 = 1, \qquad w_k = 1/k \quad (k \geq 1),$$

(2) is the strongly convergent sum $\sum w_k P_k$. Its norm is at most one. The real weights prove self-adjointness. If $\mathsf{S}f = 0$, projection gives $w_k P_k f = 0$ for every k; all weights are positive, so $f = 0$. Its range is dense: finite clock-chaos sums have preimages obtained by multiplying their finitely many components by w_k^{-1}, and these sums are dense. These arguments apply to arbitrary E-valued coefficients and retain all scenery sectors.

Lemma 4.1 (Predictable projection and shift)**.** *Clock-chaos projection on time sections preserves $\mathcal{D}_T$ and defines an orthogonal projection Π_k there. For every $u \in \mathcal{D}_T$,*

$$\mathsf{J}_T \Pi_k u = P_{k+1} \mathsf{J}_T u. \tag{20}$$

The strong series (7) *defines a contraction on $\mathcal{D}_T$ and*

$$\mathsf{J}_T \mathsf{A}_T u = N_B^{-1} \mathsf{J}_T u = \mathsf{S} \mathsf{J}_T u. \tag{21}$$

Proof. Start with scalar predictable elementary fields $1_{(s,t]}F$, where F is a polynomial of finitely many Brownian coordinates from times at most s. Its chaos projection is still a function of those past coordinates, hence is predictable. More generally conditional expectation onto the Brownian past is second quantization of the orthogonal projection onto $L^2(0,s]$; applying it to each homogeneous multiple integral commutes with taking its degree. Thus the same assertion holds for every past L^2 coefficient by polynomial density. Clock-chaos projection is a contraction in the full time-probability norm. Density of predictable elementary fields and closedness of the predictable subspace extend this action to all of $\mathcal{D}_T$. For E-valued fields use finite tensors with vectors of E, then their density; no scenery-chaos truncation is imposed.

For the elementary field, multiply its kth past chaos by the independent fresh Brownian increment. In the Gaussian product formula all contractions with that increment vanish, since its Cameron–Martin direction is orthogonal to the past. Its stochastic integral therefore has degree $k+1$. Expanding the coefficient in orthogonal degrees proves (20) on the elementary fields. Bounded projections, the Itô isometry and density extend it to every field. Sectionwise orthogonality and

completeness give $\sum \Pi_k u = u$ and $\sum \|\Pi_k u\|^2 = \|u\|^2$. Consequently (7) converges strongly and has norm at most one. Apply (20) to its finite sums and pass to the limit. The Itô integral has zero clock mean as an E-valued variable, so its P_0 component vanishes. This proves both equalities in (21). □

The predictable action is an actual section action, not an operator defined only by matching terminal stochastic integrals. Finite projection sums commute with the product-L^2 time-section isometry. Their strong convergence has an almost-everywhere convergent subsequence in time, which identifies the limiting section with the corresponding sectionwise chaos series wherever that series is already known to converge.

Applying (21) to Proposition 3.1 gives

$$\mathsf{S}F_{\varepsilon,T} = \mathsf{J}_T \mathsf{A}_T U_\varepsilon + \mathsf{S}C_{\varepsilon,T}. \tag{22}$$

This identity alone does not show that either term on the right converges: their possible cancellation has not yet been controlled.

Source correspondences:
`GaussianHorizonChaosIntertwining.horizonProjection_eq`,
`GaussianSourceHorizonRegularization.integral_sourceProjection`,
`GaussianSourceHorizonRegularization.integral_sourceRegularization`,
`GaussianPredictableSourceSlices.sourceProjection_slices`, and
`IBMRegularizedStochasticIdentity.smoothing_integral`.

5 A directional recovery estimate

Let H_B be the clock Cameron–Martin space, and let D_B be the original closed Gaussian derivative. On finite chaos expansions, D_B takes degree j to degree $j-1$ and

$$\|D_B P_j f\|^2 = j\|P_j f\|^2.$$

It follows by orthogonal summation that $D_B N_B^{-1}$ extends boundedly to all $\mathcal{H}$, with norm at most one:

$$\|D_B N_B^{-1} f\|^2 = \sum_{j\geq 1} \frac{1}{j}\|P_j f\|^2 \leq \|f\|^2. \tag{23}$$

Here its target is $L^2(\Omega_B; H_B \otimes E)$. Tensoring the scalar operator with E is justified first on finite orthogonal tensors and then by completion.

Lemma 5.1 (Rank-one graph recovery)**.** *For a unit $v \in H_B$, suppose that f is in the actual closed derivative domain and $D_B f = v \otimes G$. Then*

$$R_v f := \langle v, D_B N_B^{-1} f\rangle = \sum_{k\geq 0} \frac{1}{k+1} P_k G, \qquad \|R_v f\| \leq \|f\|. \tag{24}$$

The series converges strongly in $\mathcal{H}$.

Proof. The bound follows from (23) and contraction of a Hilbert tensor against a unit vector. The closed derivative chaos formula gives $D_B P_{k+1} f = P_k D_B f = v \otimes P_k G$. In $D_B N_B^{-1} f$ this component has coefficient $1/(k+1)$. Contracting against v proves the displayed coefficient formula. The series is strong by orthogonality and $G \in \mathcal{H}$; alternatively its partial sums converge as the projections of the bounded recovery operator. Thus the result uses the original derivative graph rather than merely proposed coefficients for G. □

At time $t > 0$ the actual clock coordinate is $B_t = \sqrt{t}\, X(v_t)$. Write the reflected endpoint as

$$F_{\varepsilon,t} = h_\varepsilon(\sqrt{t}\,|X(v_t)|) - h_\varepsilon(0).$$

It has closed clock derivative

$$D_B F_{\varepsilon,t} = \sqrt{t}\, v_t \otimes U_{\varepsilon,t}. \tag{25}$$

There is no ordinary differentiability assertion at the reflection cusp. To check the closed graph, scalarize by a continuous $\ell : E \to \mathbb{R}$. The function $x \mapsto \ell(h_\varepsilon(\sqrt{t}|x|) - h_\varepsilon(0))$ is globally Lipschitz and bounded. Its weak derivative is $\sqrt{t}\,\mathrm{sgn}(x)\ell(h'_\varepsilon(\sqrt{t}|x|))$: integrate separately on each side of zero; continuity makes the two boundary terms cancel. Smooth approximation and compact cutoffs give convergence of function and derivative in Gaussian L^2, using their uniform bounds. Lifting the one-dimensional graph to $X(v_t)$ gives (25) after every ℓ. Finite-dimensional projections in the separable value space, and closedness of the vector derivative, recover the full graph.

Set

$$q_\varepsilon(t) = t^{-1/2} R_{v_t} F_{\varepsilon,t}, \qquad q_0(t) = t^{-1/2} R_{v_t} F_{0,t} \quad (t > 0). \tag{26}$$

Equations (25) and (24) identify $q_\varepsilon(t)$ with the time section of $\mathsf{A}_T U_\varepsilon$ for almost every $t \in (0, T]$. No derivative-domain membership of $F_{0,t}$ is assumed in defining q_0: the operator R_{v_t} is already bounded on all $\mathcal{H}$.

Proposition 5.2 (The integrable singular bound)**.** *For every $t > 0$,*

$$\|q_\varepsilon(t)\|^2 \le L^2 m_1 t^{-1/2}, \qquad \|q_0(t)\|^2 \le L^2 m_1 t^{-1/2}. \tag{27}$$

For every fixed $t > 0$, $q_\varepsilon(t) \to q_0(t)$ in $\mathcal{H}$ as the positive width tends to zero, independently of the profile.

Proof. By the recovery bound, $\|q_\varepsilon(t)\|^2 \le t^{-1}\|F_{\varepsilon,t}\|^2$. Insert (16). The limit version follows in exactly the same way from the second endpoint bound. At fixed t, apply the bounded operator $t^{-1/2}R_{v_t}$ to Lemma 2.2. Both its argument at the limit and the operator itself are fixed by the original sources, so the resulting limit does not depend on a smoothing profile. □

This is the crucial improvement over estimating the raw slope by $\|\rho_\varepsilon\|_2$. It estimates the recovered slope through the endpoint energy, before integrating in time.

Source correspondences:
`GaussianDirectionalRecovery.shifted_derivative_projection`,
`IBMCompactClockGraph.tested_closed_graph`,
`IBMCompactClockRecovery.recoveredSlope_projection`,
`IBMCompactClockRecovery.recoveredSlope_uniform_energy`, and
`IBMCompactClockRecovery.recoveredSlope_continuum_tendsto`.

6 Whole-time convergence in the original predictable space

Pointwise section convergence is insufficient without a time majorant and a measurability argument. We supply both.

Proof of Theorem 1.1. First use one fixed admissible profile and the widths $\varepsilon_n = 1/(n+1)$. Each q_{ε_n} is the time-section representative of a genuine predictable L^2 field. After zero extension outside $(0, T]$, these maps are strongly measurable from time to $\mathcal{H}$. Their pointwise $\mathcal{H}$ limit on $(0, T]$ is q_0 by Proposition 5.2; the common null set needed to identify countably many section representatives

is harmless. Hence $1_{(0,T]}q_0$ is almost everywhere strongly measurable. Its squared norm is bounded by the integrable function

$$b_T(t) = 1_{(0,T]}(t)L^2 m_1 t^{-1/2}, \qquad \int_{\mathbb{R}} b_T(t)\,dt = 2L^2 m_1\sqrt{T}.$$

Consequently it is an actual Bochner time-L^2 element.

Now choose any positive widths $\delta_n \to 0$ and any fixed admissible profile. For almost every $t > 0$ the sections converge to that same $q_0(t)$. The elementary squared-norm inequality gives

$$\|q_{\delta_n}(t) - q_0(t)\|^2 \le 2\|q_{\delta_n}(t)\|^2 + 2\|q_0(t)\|^2 \le 4L^2 m_1 t^{-1/2}.$$

Dominated convergence therefore yields

$$\int_0^T \|q_{\delta_n}(t) - q_0(t)\|_{\mathcal{H}}^2\,dt \longrightarrow 0. \tag{28}$$

The zero-extension and time-section map embeds $\mathcal{D}_T$ isometrically into $L^2(\mathbb{R};\mathcal{H})$. Its image is closed, since $\mathcal{D}_T$ is complete. The canonical sequence from the first paragraph thus shows that $1_{(0,T]}q_0$ is in this image. Let $u_{0,T}$ be its unique preimage. Equation (28) proves convergence to it in the original space $\mathcal{D}_T$ for every such sequence of widths. In metric spaces the positive-width limit at zero is characterized by all positive sequences tending to zero. If that limit failed, one could choose successively $0 < \delta_n < 1/(n+1)$ whose errors stay above a fixed positive number, contradicting (28). This proves (8) for the whole continuum of widths, not just the canonical sequence.

The definition (11) is independent of the profile, so the predictable limit is independent as well. Integrating (27) gives (12). By endpoint convergence and boundedness of S, the left side of (22) converges in $\mathcal{H}$. By (8) and the Itô isometry its stochastic term also converges. Thus its actual remaining term satisfies

$$\mathsf{S}C_{\varepsilon,T} \longrightarrow c_{0,T} := \mathsf{S}F_{0,T} - \mathsf{J}_T u_{0,T} \quad \text{in } \mathcal{H}.$$

This proves (9) and (10). The residual expression defines the limit of already constructed raw corrections; it does not replace their finite-width definition (6). □

Every scenery sector is present in this argument. The convergence of sections is in $\mathcal{H}$, and the time norm in (28) integrates its full joint squared norm. A proof only after scenery expectation or only after finitely many source tests would not establish this result.
Source correspondences:
`GaussianPredictableSourceSlices.regularization_slices_of_clock_limit`,
`IBMRegularizedTimeLimit.majorant_integrable`,
`IBMRegularizedTimeLimit.exists_predictable_limit`,
`IBMDilatedPredictableLimit.slices_sequence_tendsto`,
`IBMDilatedPredictableLimit.field_continuum_tendsto`, and
`IBMDilatedPredictableLimit.correction_continuum_tendsto`.

7 The exact correction norm and fixed-sector recovery

The smoothing in (9) specifies a genuine weaker norm on the original joint variables:

$$\|f\|_- := \|\mathsf{S}f\|_{\mathcal{H}}. \tag{29}$$

Its definiteness follows from injectivity. It is not equivalent to the original norm: if e_k is a unit vector in clock chaos $k \ge 1$, then $\|e_k\|_- = 1/k$ while $\|e_k\| = 1$.

Give the actual test image $\mathcal{T} = \mathsf{S}\mathcal{H}$ the unique preimage norm $\|\mathsf{S}\psi\|_{\mathcal{T}} = \|\psi\|_{\mathcal{H}}$. This is well-defined by injectivity and makes $\mathcal{T}$ a Hilbert space. Its inclusion in $\mathcal{H}$ is continuous and dense. Self-adjointness and Cauchy–Schwarz give

$$\sup_{\|\psi\|\le 1} |\langle f, \mathsf{S}\psi\rangle| = \sup_{\|\psi\|\le 1} |\langle \mathsf{S}f, \psi\rangle| = \|f\|_{-}. \tag{30}$$

For a nonzero smoothed field the last supremum is attained at $\psi = \mathsf{S}f/\|\mathsf{S}f\|$; it is zero when $f = 0$. Thus this is the operator norm of an actual functional on a specified test space, not only a collection of individual pairings. The completion of the original space in this norm is isometric to $\mathcal{H}$ via the dense-range isometry $f \mapsto \mathsf{S}f$. Under this identification the limit correction is represented by $c_{0,T}$. It need not have an unsmoothed $\mathcal{H}$ representative. The completion observation is a direct human Hilbert-space consequence of injectivity and dense range, not an extra named formal completion construction.

Proposition 7.1 (Uniform dual convergence)**.** *With the limits of Theorem 1.1,*

$$\sup_{\|\psi\|_{\mathcal{H}}\le 1} |\langle C_{\varepsilon,T}, \mathsf{S}\psi\rangle - \langle c_{0,T}, \psi\rangle| \longrightarrow 0. \tag{31}$$

For each fixed $k \ge 0$,

$$P_k C_{\varepsilon,T} \longrightarrow w_k^{-1} P_k c_{0,T} \quad \textit{in the original } \mathcal{H}. \tag{32}$$

Proof. The expression under the supremum is $|\langle \mathsf{S}C_{\varepsilon,T} - c_{0,T}, \psi\rangle|$. Its supremum is the norm of this difference, which tends to zero by (9). For the fixed sector apply P_k to the same convergence, use $P_k\mathsf{S} = w_k P_k$, and multiply by the fixed scalar w_k^{-1}. □

The inverse factor in (32) is 1 at $k = 0$ and k at positive k. It is unbounded across sectors. Convergence of every fixed sector therefore does not justify summing the raw limits in $\mathcal{H}$.
Source correspondences:
`IBMCommonClockDual.smoothing_pairing`,
`IBMCommonClockDual.smoothing_denseRange`,
`IBMCommonClockDual.dual_attainment`,
`IBMDilatedDualClosure.correction_dual_uniform`, and
`IBMDilatedDualClosure.raw_clock_projection_tendsto`.

8 Independent rectangular source and clock tails

Let Q_m be the mth scenery-chaos projection, acting on E. It commutes with every clock projection. Set

$$R_{M,N} = \Big(\sum_{m=0}^{M} Q_m\Big)\Big(\sum_{k=0}^{N} P_k\Big).$$

These are orthogonal projections on $\mathcal{H}$. They commute with S and converge strongly to the identity as M, N tend independently to infinity. Indeed, writing $Q_{\le M}$ and $P_{\le N}$ for the two factors,

$$\|f - R_{M,N} f\| \le \|f - Q_{\le M} f\| + \|f - P_{\le N} f\|,$$

which tends to zero by completeness of the separate chaos decompositions. The two truncation parameters need not coincide.

Theorem 8.1 (Uniform full-correction tails)**.** *For every $\eta > 0$ there exist $M_0, N_0 \in \mathbb{N}$ and $\varepsilon_0 > 0$ such that for all $M \geq M_0$, $N \geq N_0$ and $0 < \varepsilon < \varepsilon_0$,*

$$\|C_{\varepsilon,T} - R_{M,N} C_{\varepsilon,T}\|_- < \eta. \tag{33}$$

For each fixed (m, k) the original joint block satisfies

$$Q_m P_k C_{\varepsilon,T} \longrightarrow w_k^{-1} Q_m P_k c_{0,T} \quad \text{in } \mathcal{H}.$$

Proof. Choose M_0, N_0 so that $\|(I - R_{M_0,N_0})c_{0,T}\| < \eta/2$. Larger rectangular projections have smaller errors, since their ranges contain this range and they are orthogonal projections. Choose ε_0 so that $\|\mathsf{S}C_{\varepsilon,T} - c_{0,T}\| < \eta/2$ for every $0 < \varepsilon < \varepsilon_0$. Commutation and contractivity give

$$\begin{aligned} \|C_{\varepsilon,T} - R_{M,N} C_{\varepsilon,T}\|_- &= \|(I - R_{M,N})\mathsf{S}C_{\varepsilon,T}\| \\ &\leq \|\mathsf{S}C_{\varepsilon,T} - c_{0,T}\| + \|(I - R_{M,N})c_{0,T}\| < \eta. \end{aligned}$$

The fixed-block assertion follows by applying the contraction Q_m to (32). □

This uniform result follows from strong convergence of the full smoothed correction. Separate convergence for every fixed block would not imply it. Equally, (33) is a dual-norm tail estimate; it asserts no uniform tail bound for the raw correction in the original norm. Both qualifiers are essential.

Source correspondences:
`IBMDilatedDualClosure.uniform_joint_tail`,
`IBMDilatedDualClosure.correction_uniform_rectangular_tail`, and
`IBMDilatedDualClosure.raw_block_tendsto`.

9 Why the clock vacuum is retained

A scalar model explains the P_0 term without appealing to the IBM limit. Let γ be standard Gaussian measure and let h_n be its orthonormal probabilists' Hermite polynomials. Then $h_n' = \sqrt{n}\, h_{n-1}$. Define

$$\mathcal{B}f = \sum_{n \geq 0} w_n \sqrt{(n+1)(n+2)}\, \langle f, h_{n+2} \rangle h_n.$$

For $n = 0$ the squared gain is 2, while for $n \geq 1$ it is $(n+1)(n+2)/n^2 \leq 6$. Orthogonality proves $\|\mathcal{B}\| \leq \sqrt{6}$; equality holds on h_3, whose image is $\sqrt{6}h_1$. If f has two successive actual closed Gaussian derivatives, differentiating its Hermite coefficients twice gives

$$\mathcal{B}f = \mathsf{S}f''.$$

This equality concerns the actual closed derivative graph. The bounded left side extends beyond that twice differentiable domain, without asserting that the ordinary second derivative has become an L^2 function there.

In particular $|x|$ has first closed derivative $\operatorname{sgn}(x)$, and

$$\langle \mathcal{B}|x|, h_0 \rangle = \mathbb{E}|G| > 0.$$

To check the last equality directly, $h_2 = (x^2 - 1)/\sqrt{2}$, so the constant coefficient is $\mathbb{E}[|G|(G^2 - 1)]$. Integration on the positive half-line gives $\mathbb{E}|G|^3 = 2\mathbb{E}|G|$, proving the claim. The ordinary second derivative away from zero is zero, but its use alone would erase this nonzero boundary channel.

With N_B^{-1} alone the clock-vacuum component of the correction is also erased. Adding P_0 makes the common correction norm definite and preserves this channel.

The scalar calculation is an illustration of the shared regularization, not an identification of the entire IBM correction with $\mathcal{B}|x|$. The joint formula also has the nonlinear scenery coefficients and all clock/source sectors. Nor does the norm $\sqrt{6}$ replace the different first-order recovery bound used in Proposition 5.2.

Source correspondences:
`GaussianRegularizedSecondDerivative.regularizedSecond_actual_graph`,
`GaussianRegularizedSecondDerivative.regularizedSecond_norm`, and
`GaussianReflectedSecondBoundary.absolute_boundary_coefficient`.

10 Classical context and the remaining mathematical boundaries

Gaussian number operators, chaos projections, Hilbert-valued Itô integration and reflected local-time formulas are classical. Maas and van Neerven [1] prove vector-valued Clark–Ocone representations in UMD spaces. Their Theorems 6.6 and 6.7 include the differentiated and boundedly extended representations; the latter's complete short proof uses density and the integral isomorphism onto its range. We directly compared those statements and their proofs, and their closed derivative/divergence setup. This is relevant prior art for value-space stochastic integration and extension by completion. Our source space $E = L^2(\Omega_W)$ is a Hilbert space, so its vector-valued nature is not a new stochastic-integration principle. We did not review every proof or hypothesis of their general UMD integration theorem.

The specific result assembled here concerns the actual reflected scenery family, its mollifier-independent recovered integrand, the integrable origin bound and the common correction topology. The bounded literature search does not certify priority for that assembly. This is a formalization companion with complete proofs of its displayed convergence mechanism and exact formal source correspondences, not a claim to a newly discovered Gaussian calculus.

Three distinctions delimit the result. First, J_T integrates against the original Brownian clock. The entire scenery remains the value-space parameter, so the integrand may use information unavailable from the observed past of Z. No theorem identifies $u_{0,T}$ as an observed-filtration IBM integrand.

Second, S is fixed and injective but has no bounded inverse on its range with the inherited $\mathcal{H}$ norm. The high-chaos vectors following (29) prove this directly. One cannot apply an uncontrolled inverse to (10) to obtain an unsmoothed L^2 Itô identity. The theorem does not prove that the raw slopes or raw corrections converge, and does not assert they diverge. The two terms in (6) are only controlled jointly after the common smoothing in the limit theorem.

Third, the convergence is strong in the entire original predictable norm for the regularized integrand, strong in the common dual norm for the full correction, and strong in ordinary joint L^2 for each fixed clock sector. These different claims cannot be substituted for one another. Neither independent rectangular dual tails nor full source retention supplies unsmoothed raw L^2 tails.

The other IBM companions [3] treat spatial grid contractions, faithful random-kernel test spaces, graph-domain obstructions, actual time averages and static intrinsic adjoints. Those statements retain their own topologies. In particular the present regularization does not remove the positive-time native clock derivative obstruction for IBM point observations. A full physical-time calculus in the observed filtration would still require a specified closed observed integral, admissible causal coefficients and a reconstruction theorem in its own topology. Those obligations remain open here.

References

[1] J. Maas and J. van Neerven, *A Clark–Ocone formula in UMD Banach spaces*, Electronic Communications in Probability **13** (2008), 151–164. Primary author manuscript, arXiv:0709.2021v2, Sections 3–4 and Theorems 6.6–6.7 (statements and the displayed proofs). `https://arxiv.org/pdf/0709.2021`.

[2] Project Stardust, *OperatorDerivativeRewrite*, frozen verified source commit given in Section 1. The accompanying `theorem-map.json` records exact headers, hashes and line links.

[3] Project Stardust, reviewed editorial companions: *Variation and spatial Simpson integration for reflected-clock iterated Brownian motion* (`../ibm-spatial-calculus/main.pdf`); *Gaussian clock regularity and domain obstructions for reflected-clock Brownian motion* (`../ibm-clock-domain/main.pdf`); *Brownian clock mixtures* (`../ibm-one-time-laws/main.pdf`); and the constructed Tanaka/holding-measure account in `../sticky-clock-recovery/main.pdf`. These are independent publication drafts from the same frozen baseline, not external priority evidence.